Software Configuration Management

Modified from Roger S. Pressman, Software Engineering:

A Practitioner's Approach 8th Edition, McGraw Hill, 2014

Introduction

- The First Law [Bersoff, et al. 1980]
  - "No matter how you are in the system lifecycle, the system will change, and the desire to change it will persist throughout the life cycle" ภายใน SDLC
- SCM is the art of identifying, organizing, and controlling modifications to the software being built by a team
- The goal is to maximize productivity by minimizing mistakes
- SCM can be applied throughout the software process, because change can occur any time

Introduction (cont.)

- SCM activities are developed to
  - identify change
  - control change
  - ensure that change is implemented
  - report change
- Software maintenance and SCM are difference
- Maintenance occur after delivery
- SCM begins when a software project begins and terminate when the software is operated

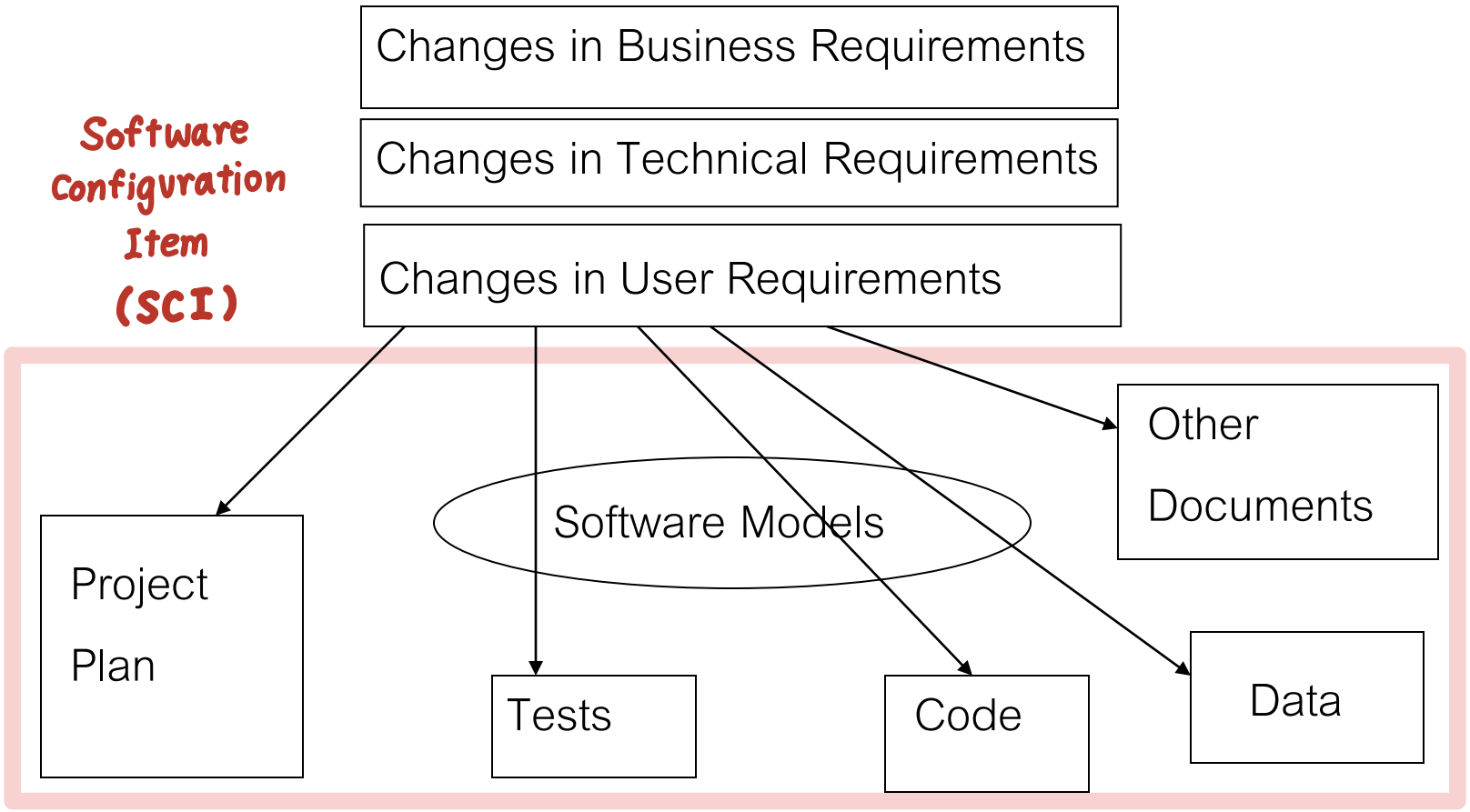
Software Configuration Management

- Output of software process may be divided into:
  - computer programs
  - documents describe the computer program
  - data
- The items produced as part of the software process are collectively called a software configuration
- Change can occur any time during the development

Sources of Changes

- new business or new market conditions
- new customer needs:
  - modification of data entered or produced by the system
  - functionality
- reorganization or business downsizing in software engineering team structure
- budgetary and scheduling constraints

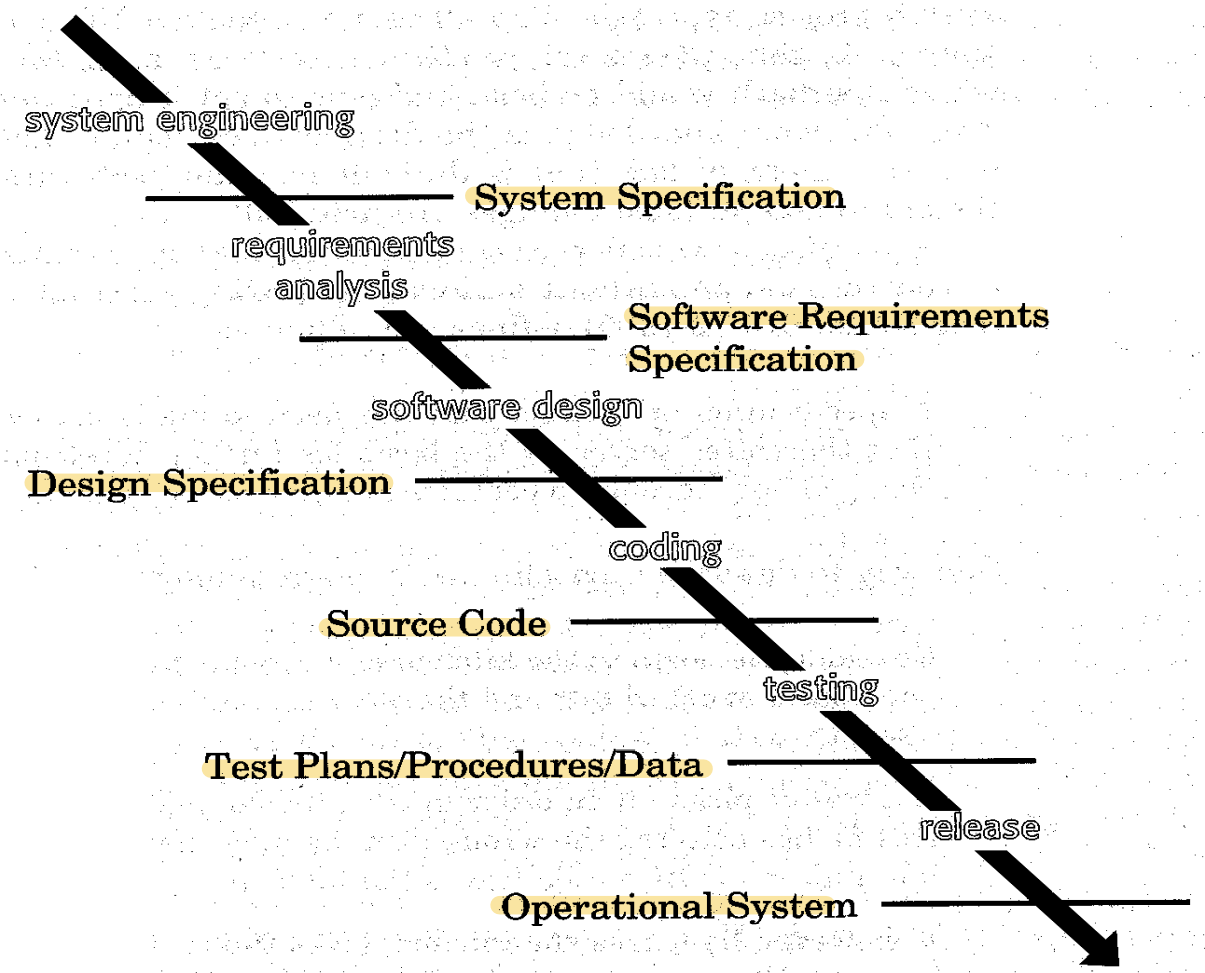
What are these Changes?



Baselines

- A baseline is a specification that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures [IEEE]
- A baseline is a milestone marked by the delivery and approved by FTR

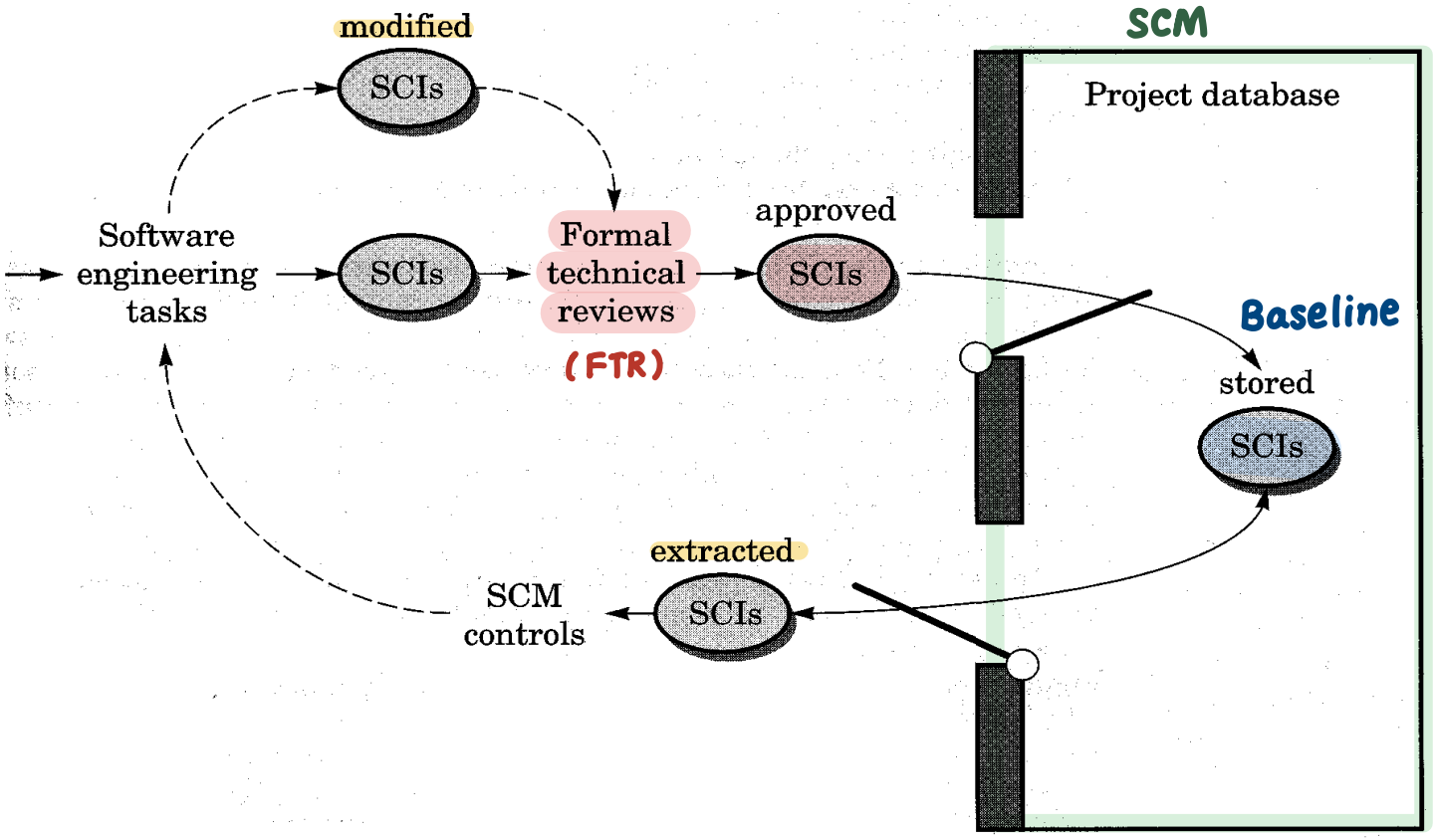
Baselines


system engineering
System Specification
requirements analysis
Software Requirements Specification
software design
Design Specification
coding
Source Code
testing
Test Plans/Procedures/Data
release
Operational System

Software Configuration Items

- A Software Configuration Items (SCI) is information created as part of software engineering process
- examples of SCI:
  - project plan
  - requirements spec
    - graphical analysis models
  - design spec: data design, architectural design, module design, interface design descriptions
  - source code
  - test specification: test plan and test procedure; test cases and recorded results; operation and user manual

Baselines SCIs



SCM Questions

- How do we manage requests for change?
- What and where are the software components?
- What is the status of each software component?
- How does a change to one component affect others?
- How do we resolve conflicting to changes?
- How do we maintain multiple versions?
- How do we keep the system up to date?

Identification of Objects in the Software Configuration

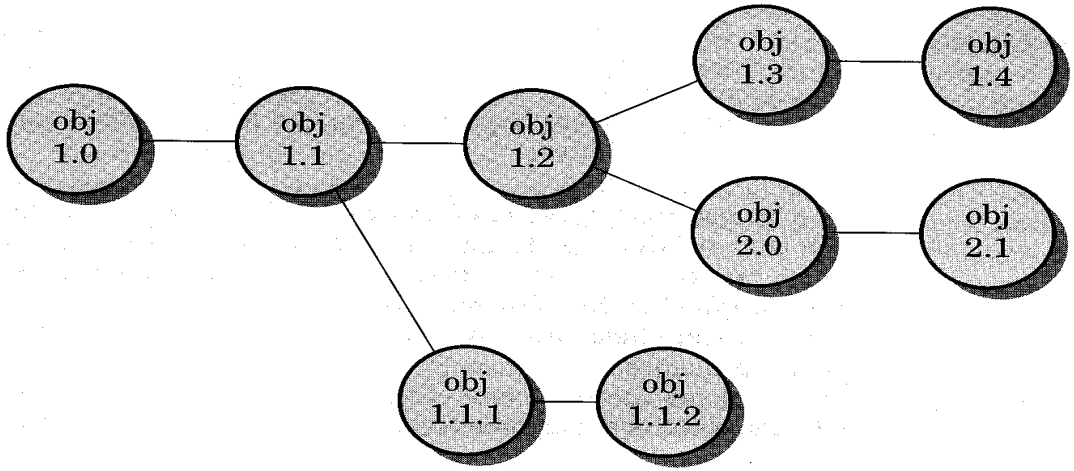
- each SCI must be identified
- There are 2 types of objects: basic and aggregate objects
- basis object is a “unit of text” that has been created in every phase
- an aggregate object is a collection of basic objects and other aggregate objects

Identification of Objects in the Software Configuration

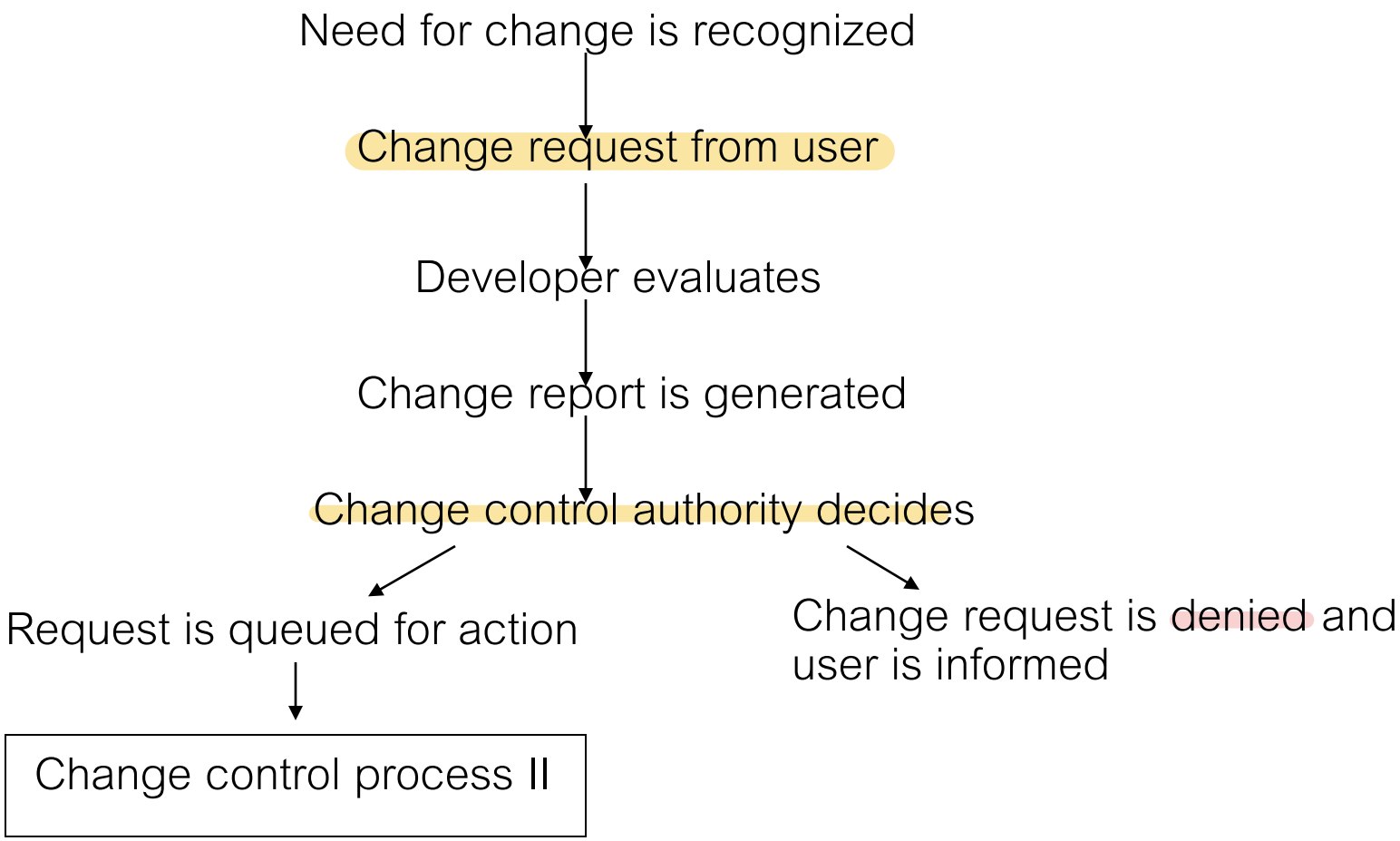
- each object is described in terms of
  - a name
  - a description that identifies the SCI type (document, program, data), project identifier; change and/or version information
  - a list of resources that are provided, processed, referenced, or required by the object

Identification of Objects in the Software Configuration (cont.)

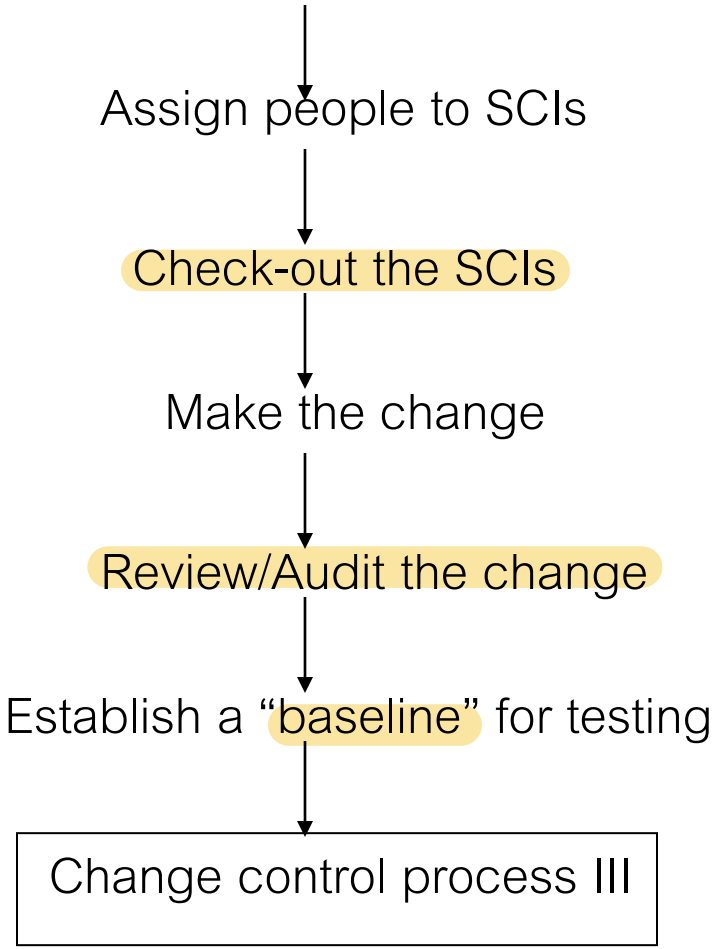
- Objects evolve throughout the software process
- an evolution graph for any object may created to show the change history of the object



Change Control Process: I



Change Control Process: II



Change Control Process: III

Perform SQA and testing activities

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Check-in the changed SCIs

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Promote SCI for inclusion in next release

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Rebuild appropriate version

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Review/Audit the change

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Include all changes in release

Software Configuration Audit

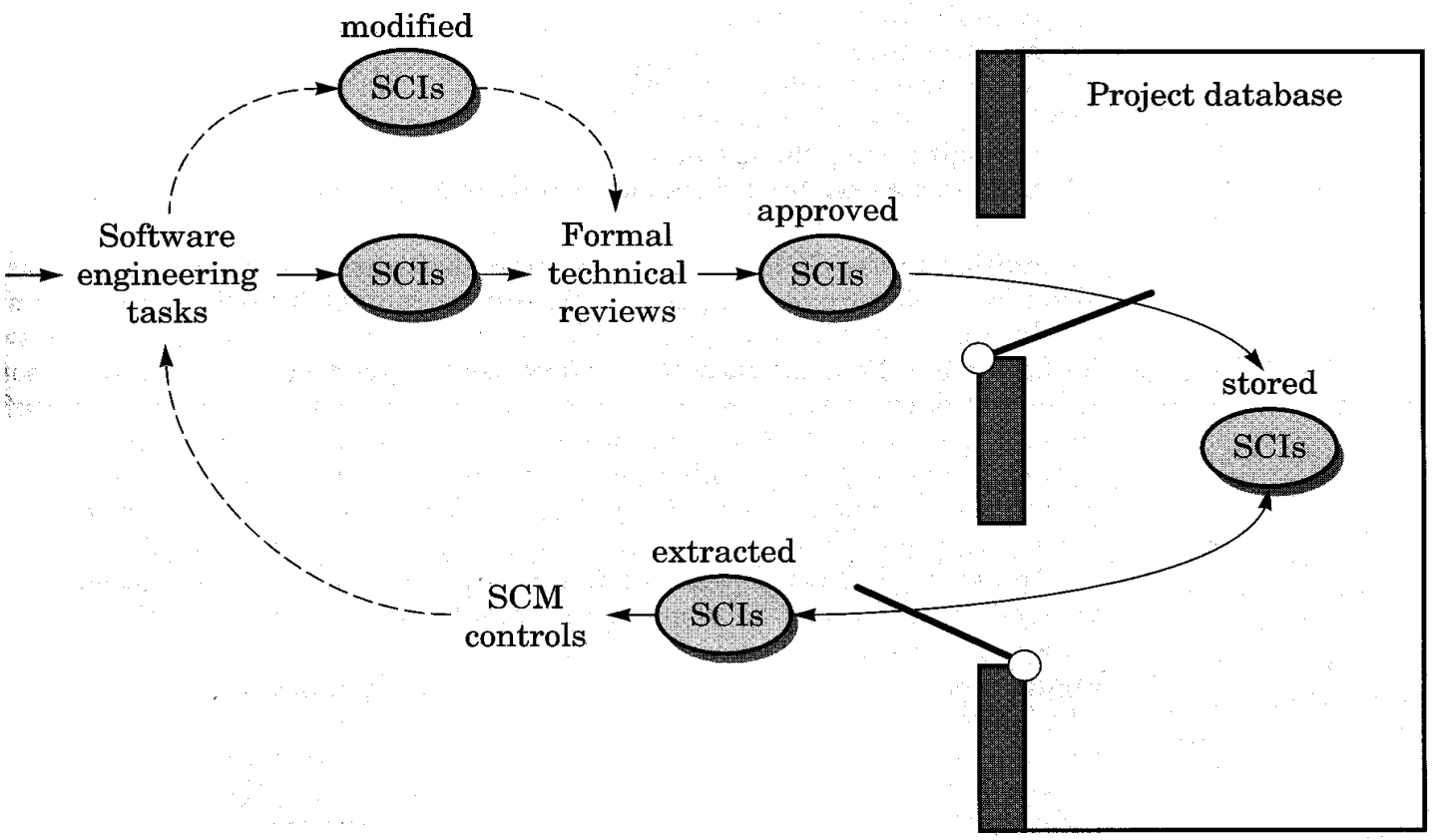
- Complements the Formal Technical Review by assessing a configuration object that are not considered during review
- Has the change been made, highlighted?
- Has a reviewed been conducted?
- Has the software process or standard been followed
- Have the change date and change author been specified?
- Have the SCM procedures for noting the change, recording it, and reporting it been followed?
- Have all related SCIs been properly updated?

Configuration Status Reporting (CSR)

- Sometimes called status accounting
- A CSR entry is made when
  - an SCI is assigned new or updated identification
  - a change is approved
  - a configuration audit is conducted
- Output from CSR may be placed in an on-line database

สรุป

- CHANGES in SW Process
- เราสนใจสิ่งไหนใน process ให้เลือกระบุมา → identification
- เรียกสิ่งนั้นว่า SCI (Software Configuration Item)
- เรามีแนวคิดในการนิยาม SCI เช่น Name, type, desc, version,…
- เรามาสนใจติดตามและควบคุม โดยมีระบบการควบคุม
  - เริ่มด้วยมี Baseline แรกเริ่มก่อน เช่น baseline SCI
  - แล้วระบบจะให้นำ Baseline เดิมไปแก้ไขเปลี่ยนแปลง และในที่สุดมีการ approve new baseline ต่อไป
  - เรามักใช้วิธีการ versioning ในการจัดการ baseline SCI เพื่อให้สามารถย้อนกลับได้


modified
SCIs
Project database
Software
engineering
tasks
SCIs
Formal
technical
reviews
approved
SCIs
stored
SCIs
extracted
SCIs
SCM
controls